Process & Decision Documentation

Entry Header

Name: Ksenia Li

Primary responsibility for this work: Adding Debug Menu to side quest

Goal of Work Session

Briefly describe what you were trying to accomplish during this phase of the assignment.

Examples: Add debug menu to W9 Example 3 provided by Karen and David

Tools, Resources, or Inputs Used

- ChatGPT 5.2

GenAI Documentation

I used GenAI to help me code a debug menu for this week's side quest exapmle

If GenAI was used (keep each response as brief as possible):

Date Used: March 23rd

Tool Disclosure: ChatGPT 5.2

Purpose of Use: debugging and coding

Summary of Interaction: By implementing instructions, GenAI generated a debug menu for me

Appendix

Please include a full transcript of your conversation with the GenAI. GenAI transcripts are used for transparency and verification only and are not assessed for writing quality or completeness.

Note: If GenAI is used in another language, you must include both the original language and a translated transcript in the appendix

add a debug screen to this game the following code is from the sketch.js: Week 9 — Example 3: Adding Sound & Music Course: GBDA302 | Instructors: Dr. Karen Cochrane & David Han Date: Mar. 19, 2026 Controls: A or D (Left / Right Arrow) Horizontal movement W (Up Arrow) Jump Space Bar Attack Tile key: g = groundTile.png (surface

```
ground) d = groundTileDeep.png (deep ground, below surface) = empty (no sprite) */ let
player; let playerImg, bgImg; let jumpSfx, musicSfx; let musicStarted = false; let
playerAnis = { idle: { row: 0, frames: 4, frameDelay: 10 }, run: { row: 1, frames: 4,
frameDelay: 3 }, jump: { row: 2, frames: 3, frameDelay: Infinity, frame: 0 }, attack: { row:
3, frames: 6, frameDelay: 2 }, }; let ground, groundDeep; let groundImg,
groundDeepImg; let attacking = false; // track if the player is attacking let
attackFrameCounter = 0; // tracking attack animation // --- TILE MAP --- // an array that
uses the tile key to create the level let level = [ " ", " ", " ", " ", " ", " ggg ",
"gggggggggggggg", // surface ground "ddddddddddddd", // deep ground ]; // --- LEVEL
CONSTANTS --- // camera view size const VIEWW = 320, VIEWH = 180; // tile width &
height const TILE_W = 24, TILE_H = 24; // size of individual animation frames const
FRAME_W = 32, FRAME_H = 32; // Y-coordinate of player start (4 tiles above the bottom)
const MAP_START_Y = VIEWH - TILE_H * 4; // gravity const GRAVITY = 10; function
preload() { // --- IMAGES --- playerImg = loadImage("assets/foxSpriteSheet.png");
bgImg = loadImage("assets/combinedBackground.png"); groundImg =
loadImage("assets/groundTile.png"); groundDeepImg =
loadImage("assets/groundTileDeep.png"); // --- SOUND --- if (typeof loadSound ===
"function") { jumpSfx = loadSound("assets/sfx/jump.wav"); musicSfx =
loadSound("assets/sfx/music.wav"); } } function setup() { // pixelated rendering with
autoscaling new Canvas(VIEWW, VIEWH, "pixelated"); // needed to correct an visual
artifacts from attempted antialiasing allSprites.pixelPerfect = true; world.gravity.y =
GRAVITY; // Try to start background music immediately. if (musicSfx)
musicSfx.setLoop(true); startMusicIfNeeded(); // --- TILE GROUPS --- ground = new
Group(); ground.physics = "static"; ground.img = groundImg; ground.tile = "g";
groundDeep = new Group(); groundDeep.physics = "static"; groundDeep.img =
groundDeepImg; groundDeep.tile = "d"; // a Tiles object creates a level based on the
level map array (defined at the beginning) new Tiles(level, 0, 0, TILE_W, TILE_H); // ---
PLAYER --- player = new Sprite(FRAME_W, MAP_START_Y, FRAME_W, FRAME_H); //
create the player player.spriteSheet = playerImg; // use the sprite sheet
player.rotationLock = true; // turn off rotations (player shouldn't rotate) // player
animation parameters player.anis.w = FRAME_W; player.anis.h = FRAME_H;
player.anis.offset.y = -4; // offset the collision box up player.addAnis(playerAnis); //
add the player animations defined earlier player.ani = "idle"; // default to the idle
animation player.w = 18; // set the width of the collsion box player.h = 20; // set the
height of the collsion box player.friction = 0; // set the friciton to 0 so we don't stick to
walls player.bounciness = 0; // set the bounciness to 0 so the player doesn't bounce //
--- GROUND SENSOR --- for use when detecting if the player is standing on the ground
sensor = new Sprite(); sensor.x = player.x; sensor.y = player.y + player.h / 2; sensor.w =
```

player.w; sensor.h = 2; sensor.mass = 0.01; sensor.removeColliders(); sensor.visible = false; let sensorJoint = new GlueJoint(player, sensor); sensorJoint.visible = false; } function startMusicIfNeeded() { if (musicStarted || !musicSfx) return; const startLoop = () => { if (!musicSfx.isPlaying()) musicSfx.play(); musicStarted = musicSfx.isPlaying(); }; // Some browsers require a user gesture before audio can start. const maybePromise = userStartAudio(); if (maybePromise && typeof maybePromise.then === "function") { maybePromise.then(startLoop).catch(() => {}); } else { startLoop(); } } function keyPressed() { startMusicIfNeeded(); } function mousePressed() { startMusicIfNeeded(); } function touchStarted() { startMusicIfNeeded(); return false; } function draw() { // --- BACKGROUND --- camera.off(); imageMode(CORNER); image(bgImg, 0, 0, bgImg.width, bgImg.height); camera.on(); // --- PLAYER CONTROLS --- // first check to see if the player is on the ground let grounded = sensor.overlapping(ground); // -- ATTACK INPUT -- if (grounded && !attacking && kb.presses("space")) { attacking = true; attackFrameCounter = 0; player.vel.x = 0; player.ani.frame = 0; player.ani = "attack"; player.ani.play(); // plays once to end } // -- JUMP -- if (grounded && kb.presses("up")) { player.vel.y = -4; if (jumpSfx) jumpSfx.play(); } // --- STATE MACHINE --- if (attacking) { attackFrameCounter++; // Attack lasts ~6 frames * frameDelay 2 = 12 cycles (adjust if needed) if (attackFrameCounter > 12) { attacking = false; attackFrameCounter = 0; } } else if (!grounded) { player.ani = "jump"; player.ani.frame = player.vel.y < 0 ? 0 : 1; } else { player.ani = kb.pressing("left") || kb.pressing("right") ? "run" : "idle"; } // --- MOVEMENT --- if (!attacking) { player.vel.x = 0; if (kb.pressing("left")) { player.vel.x = -1.5; player.mirror.x = true; } else if (kb.pressing("right")) { player.vel.x = 1.5; player.mirror.x = false; } } // --- KEEP IN VIEW --- player.pos.x = constrain(player.pos.x, FRAME_W / 2, VIEWW - FRAME_W / 2); }

Thought for 5s

Yes — for this side quest, your debug screen should include **moon gravity on/off** plus **at least one more useful dev toggle**. That is exactly what page 18 asks for, and page 4 also frames the debug menu as a way to inspect game state and tune values quickly.

Use this directly in your `sketch.js`.

What this adds

- Press ` (backtick) to show/hide the debug menu
- Press **1** to toggle **moon gravity**
- Press **2** to toggle **show collision/sensor boxes**
- Press **3** to toggle **player invincibility** as the second debug feature
- On-screen panel showing current values

1) Add these variables near the top of `sketch.js`

Put them under your existing globals:

```
let player;
let playerImg, bgImg;
let jumpSfx, musicSfx;
let musicStarted = false;

let playerAnis = {
  idle: { row: 0, frames: 4, frameDelay: 10 },
  run: { row: 1, frames: 4, frameDelay: 3 },
  jump: { row: 2, frames: 3, frameDelay: Infinity, frame: 0 },
  attack: { row: 3, frames: 6, frameDelay: 2 },
};

let ground, groundDeep;
let groundImg, groundDeepImg;

let attacking = false;
let attackFrameCounter = 0;

// --- DEBUG ---
let debugMenuOpen = false;
let moonGravity = false;
let showDebugBoxes = false;
let playerInvincible = false;

const NORMAL_GRAVITY = 10;
const MOON_GRAVITY = 2.2;
```

---

2) Change your gravity constant

Right now you have:

```
const GRAVITY = 10;
```

Replace that with:

```
const GRAVITY = NORMAL_GRAVITY;
```

Or, even cleaner, just remove `const GRAVITY = 10;` completely and use `NORMAL_GRAVITY` directly.

---

3) Update `setup()`

Replace this line:

```
world.gravity.y = GRAVITY;
```

with:

```
world.gravity.y = NORMAL_GRAVITY;
```

Also add debug visibility setup inside `setup()` after you create the player and sensor:

```
  player.debug = false;
  sensor.debug = false;
```

So the end of your `setup()` will look like this:

```
function setup() {
  new Canvas(VIEWW, VIEWH, "pixelated");
  allSprites.pixelPerfect = true;

  world.gravity.y = NORMAL_GRAVITY;

  if (musicSfx) musicSfx.setLoop(true);
  startMusicIfNeeded();

  ground = new Group();
  ground.physics = "static";
  ground.img = groundImg;
  ground.tile = "g";

  groundDeep = new Group();
  groundDeep.physics = "static";
  groundDeep.img = groundDeepImg;
  groundDeep.tile = "d";

  new Tiles(level, 0, 0, TILE_W, TILE_H);

  player = new Sprite(FRAME_W, MAP_START_Y, FRAME_W, FRAME_H);
  player.spriteSheet = playerImg;
  player.rotationLock = true;

  player.anis.w = FRAME_W;
  player.anis.h = FRAME_H;
  player.anis.offset.y = -4;
  player.addAnis(playerAnis);
  player.ani = "idle";
  player.w = 18;
  player.h = 20;
  player.friction = 0;
  player.bounciness = 0;

  sensor = new Sprite();
```

```
  sensor.x = player.x;
  sensor.y = player.y + player.h / 2;
  sensor.w = player.w;
  sensor.h = 2;
  sensor.mass = 0.01;
  sensor.removeColliders();
  sensor.visible = false;

  let sensorJoint = new GlueJoint(player, sensor);
  sensorJoint.visible = false;

  // --- DEBUG VISUALS ---
  player.debug = false;
  sensor.debug = false;
}
```

---

4) Replace your `keyPressed()` with this

This keeps your music startup and adds debug toggles:

```
function keyPressed() {
  startMusicIfNeeded();

  // Toggle debug menu with backtick `
  if (key === "`") {
    debugMenuOpen = !debugMenuOpen;
  }

  // Only allow debug toggles when menu is open
  if (debugMenuOpen) {
    if (key === "1") {
      moonGravity = !moonGravity;
      world.gravity.y = moonGravity ? MOON_GRAVITY : NORMAL_GRAVITY;
    }

    if (key === "2") {
      showDebugBoxes = !showDebugBoxes;
      player.debug = showDebugBoxes;
      sensor.debug = showDebugBoxes;
    }

    if (key === "3") {
      playerInvincible = !playerInvincible;
    }
  }
}
```

---

5) Add these helper functions below `touchStarted()`

Paste these before `draw()`:

```
function drawDebugMenu() {
  camera.off();

  push();
  fill(0, 180);
  noStroke();
  rect(14, 14, 200, 108, 8);

  fill(255);
  textAlign(LEFT, TOP);
  textSize(12);
  text("DEBUG MENU", 24, 24);

  textSize(10);
  text("`  Toggle Menu", 24, 44);
  text("1  Moon Gravity: " + (moonGravity ? "ON" : "OFF"), 24, 60);
  text("2  Show Hitboxes: " + (showDebugBoxes ? "ON" : "OFF"), 24, 76);
  text("3  Invincibility: " + (playerInvincible ? "ON" : "OFF"), 24, 92);
  pop();

  camera.on();
}

function drawDebugStats(grounded) {
  camera.off();

  push();
  fill(255);
  textAlign(LEFT, TOP);
  textSize(10);
  text("x: " + nf(player.x, 1, 1), 10, VIEWH - 42);
  text("y: " + nf(player.y, 1, 1), 10, VIEWH - 32);
  text("vx: " + nf(player.vel.x, 1, 2), 10, VIEWH - 22);
  text("vy: " + nf(player.vel.y, 1, 2), 10, VIEWH - 12);

  text("grounded: " + grounded, 95, VIEWH - 42);
  text("attacking: " + attacking, 95, VIEWH - 32);
  text("gravity: " + world.gravity.y, 95, VIEWH - 22);
  text("ani: " + player.ani.name, 95, VIEWH - 12);
  pop();

  camera.on();
}
```

---

6) Update your `draw()` function

Use this version of `draw()`:

```
function draw() {
  // --- BACKGROUND ---
  camera.off();
  imageMode(CORNER);
  image(bgImg, 0, 0, bgImg.width, bgImg.height);
  camera.on();
```

```
  // --- PLAYER CONTROLS ---
  let grounded = sensor.overlapping(ground);

  // Keep debug visuals synced
  player.debug = showDebugBoxes;
  sensor.debug = showDebugBoxes;

  // -- ATTACK INPUT --
  if (grounded && !attacking && kb.presses("space")) {
    attacking = true;
    attackFrameCounter = 0;
    player.vel.x = 0;
    player.ani.frame = 0;
    player.ani = "attack";
    player.ani.play();
  }

  // -- JUMP --
  if (grounded && kb.presses("up")) {
    player.vel.y = -4;
    if (jumpSfx) jumpSfx.play();
  }

  // --- STATE MACHINE ---
  if (attacking) {
    attackFrameCounter++;
    if (attackFrameCounter > 12) {
      attacking = false;
      attackFrameCounter = 0;
    }
  } else if (!grounded) {
    player.ani = "jump";
    player.ani.frame = player.vel.y < 0 ? 0 : 1;
  } else {
    player.ani = kb.pressing("left") || kb.pressing("right") ? "run" :
"idle";
  }

  // --- MOVEMENT ---
  if (!attacking) {
    player.vel.x = 0;
    if (kb.pressing("left")) {
      player.vel.x = -1.5;
      player.mirror.x = true;
    } else if (kb.pressing("right")) {
      player.vel.x = 1.5;
      player.mirror.x = false;
    }
  }

  // --- KEEP IN VIEW ---
  player.pos.x = constrain(player.pos.x, FRAME_W / 2, VIEWW - FRAME_W / 2);

  // Optional example of using invincibility later:
  // if (!playerInvincible) {
  //   // apply damage logic here
```

```
  // }

  // --- DEBUG UI ---
  if (debugMenuOpen) {
    drawDebugMenu();
    drawDebugStats(grounded);
  }
}
```

---

7) What to say if your prof/TA asks what you added

You can describe it like this:

I added a toggleable debug screen to Example 3. It includes moon gravity on/off, collision box visibility, and an invincibility toggle. The menu also displays live player state values like position, velocity, grounded state, animation state, and current gravity. This matches the purpose of the Week 9 debug example: making game systems visible for faster iteration and testing.

---

8) Tiny fix

You are using `sensor` but it is not declared at the top. Add this near your other globals:

```
let sensor;
```

Full feature checklist

This should satisfy the side quest:

- Debug screen added
- Moon gravity toggle added
- At least one additional useful feature added
- Real-time game state visible on screen

That aligns with the Week 9 requirement for Example 3.

If you want, I can also give you a **clean full final `sketch.js` pasted together in one block** so you can copy it all at once.

yes please

Absolutely — here is a **full `sketch.js`** with the debug screen added.

```
/*
  Week 9 — Example 3: Adding Sound & Music

  Course: GBDA302 | Instructors: Dr. Karen Cochrane & David Han
  Date: Mar. 19, 2026

  Controls:
    A or D (Left / Right Arrow)   Horizontal movement
    W (Up Arrow)                  Jump
    Space Bar                     Attack

  Debug Controls:
    `                              Toggle debug menu
    1                              Toggle moon gravity
    2                              Toggle hitboxes
    3                              Toggle invincibility

  Tile key:
    g = groundTile.png       (surface ground)
    d = groundTileDeep.png   (deep ground, below surface)
      = empty (no sprite)
*/

let player;
let sensor;
let playerImg, bgImg;
let jumpSfx, musicSfx;
let musicStarted = false;

let playerAnis = {
  idle: { row: 0, frames: 4, frameDelay: 10 },
  run: { row: 1, frames: 4, frameDelay: 3 },
  jump: { row: 2, frames: 3, frameDelay: Infinity, frame: 0 },
  attack: { row: 3, frames: 6, frameDelay: 2 },
};

let ground, groundDeep;
let groundImg, groundDeepImg;

let attacking = false;
let attackFrameCounter = 0;

// --- DEBUG ---
let debugMenuOpen = false;
let moonGravity = false;
let showDebugBoxes = false;
let playerInvincible = false;

const NORMAL_GRAVITY = 10;
const MOON_GRAVITY = 2.2;

// --- TILE MAP ---
let level = [
  "              ",
  "              ",
  "              ",
  "              ",
```

```
  "              ",
  "       ggg    ",
  "gggggggggggggg",
  "dddddddddddddd",
];

// --- LEVEL CONSTANTS ---
const VIEWW = 320,
  VIEWH = 180;

const TILE_W = 24,
  TILE_H = 24;

const FRAME_W = 32,
  FRAME_H = 32;

const MAP_START_Y = VIEWH - TILE_H * 4;

function preload() {
  // --- IMAGES ---
  playerImg = loadImage("assets/foxSpriteSheet.png");
  bgImg = loadImage("assets/combinedBackground.png");
  groundImg = loadImage("assets/groundTile.png");
  groundDeepImg = loadImage("assets/groundTileDeep.png");

  // --- SOUND ---
  if (typeof loadSound === "function") {
    jumpSfx = loadSound("assets/sfx/jump.wav");
    musicSfx = loadSound("assets/sfx/music.wav");
  }
}

function setup() {
  // pixelated rendering with autoscaling
  new Canvas(VIEWW, VIEWH, "pixelated");

  // needed to correct visual artifacts from attempted antialiasing
  allSprites.pixelPerfect = true;

  world.gravity.y = NORMAL_GRAVITY;

  // Try to start background music immediately
  if (musicSfx) musicSfx.setLoop(true);
  startMusicIfNeeded();

  // --- TILE GROUPS ---
  ground = new Group();
  ground.physics = "static";
  ground.img = groundImg;
  ground.tile = "g";

  groundDeep = new Group();
  groundDeep.physics = "static";
  groundDeep.img = groundDeepImg;
  groundDeep.tile = "d";

  // Create level from tile map
```

```
  new Tiles(level, 0, 0, TILE_W, TILE_H);

  // --- PLAYER ---
  player = new Sprite(FRAME_W, MAP_START_Y, FRAME_W, FRAME_H);
  player.spriteSheet = playerImg;
  player.rotationLock = true;

  // Player animation parameters
  player.anis.w = FRAME_W;
  player.anis.h = FRAME_H;
  player.anis.offset.y = -4;
  player.addAnis(playerAnis);
  player.ani = "idle";

  // Collision box
  player.w = 18;
  player.h = 20;
  player.friction = 0;
  player.bounciness = 0;

  // --- GROUND SENSOR ---
  sensor = new Sprite();
  sensor.x = player.x;
  sensor.y = player.y + player.h / 2;
  sensor.w = player.w;
  sensor.h = 2;
  sensor.mass = 0.01;
  sensor.removeColliders();
  sensor.visible = false;

  let sensorJoint = new GlueJoint(player, sensor);
  sensorJoint.visible = false;

  // --- DEBUG VISUALS ---
  player.debug = false;
  sensor.debug = false;
}

function startMusicIfNeeded() {
  if (musicStarted || !musicSfx) return;

  const startLoop = () => {
    if (!musicSfx.isPlaying()) musicSfx.play();
    musicStarted = musicSfx.isPlaying();
  };

  const maybePromise = userStartAudio();
  if (maybePromise && typeof maybePromise.then === "function") {
    maybePromise.then(startLoop).catch(() => {});
  } else {
    startLoop();
  }
}

function keyPressed() {
  startMusicIfNeeded();
```

```
  // Toggle debug menu
  if (key === "`") {
    debugMenuOpen = !debugMenuOpen;
  }

  if (debugMenuOpen) {
    if (key === "1") {
      moonGravity = !moonGravity;
      world.gravity.y = moonGravity ? MOON_GRAVITY : NORMAL_GRAVITY;
    }

    if (key === "2") {
      showDebugBoxes = !showDebugBoxes;
      player.debug = showDebugBoxes;
      sensor.debug = showDebugBoxes;
    }

    if (key === "3") {
      playerInvincible = !playerInvincible;
    }
  }
}

function mousePressed() {
  startMusicIfNeeded();
}

function touchStarted() {
  startMusicIfNeeded();
  return false;
}

function drawDebugMenu() {
  camera.off();

  push();
  fill(0, 180);
  noStroke();
  rect(14, 14, 210, 112, 8);

  fill(255);
  textAlign(LEFT, TOP);
  textSize(12);
  text("DEBUG MENU", 24, 24);

  textSize(10);
  text("`  Toggle Menu", 24, 44);
  text("1  Moon Gravity: " + (moonGravity ? "ON" : "OFF"), 24, 60);
  text("2  Show Hitboxes: " + (showDebugBoxes ? "ON" : "OFF"), 24, 76);
  text("3  Invincibility: " + (playerInvincible ? "ON" : "OFF"), 24, 92);
  pop();

  camera.on();
}

function drawDebugStats(grounded) {
  camera.off();
```

```
  push();
  fill(255);
  textAlign(LEFT, TOP);
  textSize(10);

  text("x: " + nf(player.x, 1, 1), 10, VIEWH - 42);
  text("y: " + nf(player.y, 1, 1), 10, VIEWH - 32);
  text("vx: " + nf(player.vel.x, 1, 2), 10, VIEWH - 22);
  text("vy: " + nf(player.vel.y, 1, 2), 10, VIEWH - 12);

  text("grounded: " + grounded, 100, VIEWH - 42);
  text("attacking: " + attacking, 100, VIEWH - 32);
  text("gravity: " + world.gravity.y, 100, VIEWH - 22);

  let currentAni = typeof player.ani === "string" ? player.ani :
player.ani.name;
  text("ani: " + currentAni, 100, VIEWH - 12);

  pop();

  camera.on();
}

function draw() {
  // --- BACKGROUND ---
  camera.off();
  imageMode(CORNER);
  image(bgImg, 0, 0, bgImg.width, bgImg.height);
  camera.on();

  // --- PLAYER CONTROLS ---
  let grounded = sensor.overlapping(ground);

  // Keep debug visuals synced
  player.debug = showDebugBoxes;
  sensor.debug = showDebugBoxes;

  // -- ATTACK INPUT --
  if (grounded && !attacking && kb.presses("space")) {
    attacking = true;
    attackFrameCounter = 0;
    player.vel.x = 0;
    player.ani.frame = 0;
    player.ani = "attack";
    player.ani.play();
  }

  // -- JUMP --
  if (grounded && kb.presses("up")) {
    player.vel.y = -4;
    if (jumpSfx) jumpSfx.play();
  }

  // --- STATE MACHINE ---
  if (attacking) {
    attackFrameCounter++;
```

```
    if (attackFrameCounter > 12) {
      attacking = false;
      attackFrameCounter = 0;
    }
  } else if (!grounded) {
    player.ani = "jump";
    player.ani.frame = player.vel.y < 0 ? 0 : 1;
  } else {
    player.ani = kb.pressing("left") || kb.pressing("right") ? "run" :
"idle";
  }

  // --- MOVEMENT ---
  if (!attacking) {
    player.vel.x = 0;

    if (kb.pressing("left") || kb.pressing("a")) {
      player.vel.x = -1.5;
      player.mirror.x = true;
    } else if (kb.pressing("right") || kb.pressing("d")) {
      player.vel.x = 1.5;
      player.mirror.x = false;
    }
  }

  // --- KEEP IN VIEW ---
  player.pos.x = constrain(player.pos.x, FRAME_W / 2, VIEWW - FRAME_W / 2);

  // Example placeholder for future damage logic:
  // if (!playerInvincible) {
  //   // apply damage here
  // }

  // --- DEBUG UI ---
  if (debugMenuOpen) {
    drawDebugMenu();
    drawDebugStats(grounded);
  }
}
```

Two small notes:

1. I added `A` and `D` movement support too, because your header says “A or D” but your original movement code only checked left/right arrows.
2. The debug menu is intentionally simple and assignment-safe: it does not change your main game logic except for gravity and debug toggles.

If the game throws an error after pasting this, send me your `index.html` too, because the most likely issue would be whether your project is using the right p5play/p5.sound setup.

How do i save this repository to my personal github so I can also change the file

Good question — this is exactly what you should be doing (version control + backup). I'll give you the **clean, correct workflow** used in real projects.

✅ OPTION 1 (Recommended): Upload your project to GitHub

Step 1 — Create a new repo on GitHub

1. Go to: https://github.com
2. Click **"New repository"**
3. Name it something like:

   ```
   gbda302-week9-game
   ```

4. ❗ DO NOT check:
   - README
   - .gitignore
5. Click **Create repository**

Step 2 — Copy your repo URL

You'll see something like:

```
https://github.com/yourname/gbda302-week9-game.git
```

Copy it.

Step 3 — Initialize Git in VS Code

Open **Terminal in VS Code**:

👉 Top menu:

```
Terminal → New Terminal
```

Then run:

```
git init
git add .
git commit -m "Initial commit - Week 9 debug game"
```

Step 4 — Connect to GitHub

Paste your repo URL here:

```
git remote add origin https://github.com/YOUR_USERNAME/gbda302-week9-game.git
```

Then:

```
git branch -M main
git push -u origin main
```

🔐 If it asks for login

Use:

- Username = your GitHub username
- Password = **GitHub Personal Access Token (NOT your password)**

If you don't have one, tell me — I'll walk you through it in 30 seconds.

✅ OPTION 2 (Easier UI method — no terminal)

If you want the simplest way:

In VS Code:

1. Click **Source Control icon (branch icon on left sidebar)**
2. Click **"Initialize Repository"**
3. Click **"Publish to GitHub"**
4. Choose:
    - Private or Public
5. Done ✅

🔁 How to update files later

Every time you change code:

```
git add .
git commit -m "updated debug menu"
git push
```

⚠️ IMPORTANT (for your project)

Make sure your repo includes:

```
index.html
sketch.js
style.css
/assets
/libraries
```

👉 If assets are missing → game will break on GitHub

🧠 Pro tip (this will impress in class/interviews)

Rename your repo like this:

```
p5-platformer-debug-tools
```

And add a short README:

Includes debug menu with gravity toggle, hitbox visualization, and live state inspection.

🚀 If you want next level (optional)

I can help you:

- Deploy it live (GitHub Pages → playable link)
- Clean repo structure for portfolio
- Add README with screenshots (A+ level submission)

If you get stuck on ANY step, send me a screenshot — I'll fix it with you quickly.

github is not letting me save the repositort, i tried to create a new on and manually import code but it would show a blank screen. how do i save this and rename it in another way

You're very close — this is a **common Mac permission issue**, not a GitHub problem.

The error:

```
Permission denied .git
```

means VS Code **doesn't have permission to write inside that folder** (`Downloads/updated 2/...`).

✅ FASTEST FIX (recommended)

🔧 Step 1 — Move your project out of Downloads

Downloads folders often cause permission issues.

👉 Do this:

1. Open **Finder**
2. Drag your folder:

   ```
   Week9_Example03
   ```

   into:

   ```
   Documents
   ```

   or create a new folder:

   ```
   Documents/Code/
   ```

🔧 Step 2 — Reopen in VS Code

1. Close VS Code
2. Reopen it
3. File → Open Folder
4. Select the new location:

   `Documents/Code/Week9_Example03`

🔧 Step 3 — Try again

Now click:
👉 **"Publish to GitHub"**

This should work immediately.

🧠 WHY this happened

Mac sometimes blocks:

- Downloads folder
- Folders with spaces like `updated 2`
- Files copied from the internet

Git tries to create `.git` → Mac says ❌

🚀 BACKUP METHOD (if UI still fails)

Use terminal (this always works):

In VS Code Terminal:

```
cd ~/Documents/Code/Week9_Example03

git init
git add .
git commit -m "week 9 debug game"
```

Then:

```
git remote add origin https://github.com/YOUR_USERNAME/YOUR_REPO.git
git branch -M main
git push -u origin main
```

⚠ About your “blank screen” issue

That’s separate from Git — it usually means:

❌ Missing files in repo

Make sure these exist:

```
index.html
sketch.js
/assets/
libraries/
```

If you uploaded only `.js` → page = blank

🔥 If you want ZERO friction (best option)

Do this:

👉 Use VS Code button

1. Click **“Initialize Repository”**
2. Then click **“Publish to GitHub”**

No terminal needed.

💡 Rename your repo (clean version)

Instead of:

```
Week9_Example03
```

Use:

```
p5-debug-platformer
```

Looks much more professional.

✅ Summary

Your issue = **folder permissions**, not GitHub

✔ Move project out of Downloads
✔ Reopen in VS Code
✔ Publish again

If it STILL fails, send me:

- what happens after moving folders
- or your repo link

I'll debug it with you step-by-step 👍